A Deep Reinforcement Learning-Based Grid Management Framework for PV-Rich Smart Distribution Systems

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***Abstract*—This article presents a multi-stage deep reinforcement learning framework for volt-var or volt-watt optimization (VVO/VWO) in solar photovoltaic-rich active distribution networks (ADNs) to enhance voltage regulation and minimize network losses. The framework introduces a centralized–decentralized training and execution structure for a multi-agent environment that segregates the objectives—voltage regulation and loss minimization into decentralized and centralized tasks, respectively. This architecture leverages independent learning to reduce training iterations and the computation overheads, particularly in large-scale agent deployments, while sharing limited information during centralized training. A novel contribution factor method based on reward backstepping is proposed to address the non-stationarity caused by independent learning, which enables accurate credit assignment and stable learning. The proposed framework has been evaluated and validated on a practical ADN, a modified IEEE 33 bus, and an IEEE 119 bus system that are segmented into different areas of control (AoC), where each consumer within the AoC comprises local loads and an individual PV-fed inverter, which serves as the primary controlling device for executing VVO/VWO. The results indicate effective reduction of voltage violations and network losses, improved training convergence, and decreased computational requirements compared to existing frameworks.**

***Index Terms*—Actor-critic methods, deep reinforcement learning, markov decision process, multi-agent systems, voltage regulation.**

Nomenclature

k	Index for consumer.
t	Index for time block.
g/d	Index for generation/ demand, respectively.
A_i	Index for "i^{th}" area of control (AoC).
h	Index for DRL trajectory length.
n_c	Number of consumers within any AoC.
$P^g_{k,t}/P^d_{k,t}$	Actual active power generation/ demand of consumer "k" at time "t".
$\mathbf{Q}^g_{k,t}/\mathbf{Q}^d_{k,t}$	Actual reactive power generation/ demand of consumer "k" at time "t".
$P^{g_0}_{k,t}$	Actual active power generation of consumer "k" at time "t" without manipulation.
V^{s,A_i}_t	Slack bus voltage of AoC "A_i" at time "t".
$V_{k,t}$	Bus voltage of consumer "k" at time "t".
$\mathbf{a_{k,t}}/\mathbf{b_{k,t}}$	Tangent of power factor angle and active power curtailment factor of consumer "k" at time "t".
$P^{l,D}_{cu,t}/P^{l,D}_{nl,t}$	Copper and no-load loss of the DTR at time "t".
$vio_{j,t}$	Variable representing if node "j" violates the voltage constraints at time "t".
P^{l,A_i}_t	Network loss of "i^{th}" AoC at time "t".
δ, Θ	Voltage angle and impedance angle, respectively.
$\phi_{v,k}, \phi'_{v,k}$	Parameters of main critic and target critic network respectively of consumer "k" considering individual objective.
ϕ_l, ϕ'_l	Parameters of main critic and target critic network respectively considering global goal.
θ_k, θ'_k	Parameters of main actor and target actor network respectively of consumer "k".

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I. Introduction

A. Background and Motivation

THE electrical distribution networks have witnessed a significant rise in the integration of renewable energy sources (RES), particularly solar photovoltaic (PV), wind, and hydroelectric power. The global share of RESs increased from 18% in 2005 to 34.5% in 2025 and is expected to surpass 40% by 2028, with solar PV accounting for 35.3% of total renewable capacity [1]. In India, solar PV systems account for around 63% of the total installed renewable capacity, comprising 76% of ground-mounted systems, 17% of grid-connected rooftop installations, and 7% of off-grid and hybrid projects [2]. The substantial integration of RESs have supported exponentially growing demand while, mitigating environmental impacts and reliance on fossil fuels; however, the inherent intermittency of solar PV poses considerable operational challenges. Rapid fluctuations in solar PV output may result in voltage violations, reverse power flow, increased losses, feeder congestion, miscoordination of protective systems, and excessive operations of on-load tap changers. Therefore, these uncertainties require sophisticated control strategies for reliable system operation. Volt–VAr optimization (VVO) is widely used to address these challenges. Conventional VVO relies on mechanically switched equipment, including capacitor banks, voltage regulators, OLTCs, and static

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VAR compensators, which exhibit slower response times, discrete control actions, and higher capital and maintenance costs. Conversely, intelligent PV inverters provide rapid, continuous reactive power management for voltage regulation and have become viable options under IEEE 1547.8 [3]. Model-based VVO is typically expressed as a nonlinear optimization problem. Centralized OPF-based approaches have been proposed to minimize network losses and DG curtailment [4], while accelerated ADMM-based distributed optimization enables coordinated reactive power management across ADN clusters while ensuring data privacy [5]. Decentralized VVO systems that utilize PV inverters for local voltage regulation have also been reported [6]. Furthermore, bilayer gradient projection-based VVC for radial wind farms [7] and containment-based distributed voltage control for interconnected microgrids [8] are discussed. A localized, model-free, adaptive voltage control technique utilizing extremum-seeking droop control was presented in [9], while Zhang et al. [10] introduced an inverter-based volt–var and volt–watt optimization using droop characteristics to minimize network losses and local voltage violations.

Despite their effectiveness, traditional VVO techniques face limitations in ADNs. They require accurate and continuously updated network models and parameter information, which are difficult to obtain and maintain in evolving systems. Moreover, these approaches generally require resolving the optimization problem whenever operating conditions change, delaying real-time decision-making. Further, node-level control in large participating areas also remains largely unexplored due to growth in system complexity with network size and the number of controllable variables.

B. Related Work

Data-driven optimization has emerged as a potential alternative to classical optimization methods, with deep reinforcement learning (DRL) algorithms widely used for VVO in ADNs [11], [12], [13], [14], [15]. Liu and Wu [11] developed a centralized DRL–JASAC framework to minimize modeling errors and voltage violations. A bi-level, two-timescale DRL approach in [12] enabled model-free VVO by co-optimizing OLTCs, step voltage regulators, and fast inverter-based DERs settings. Sun and Qiu [13] proposed a two-stage MA-DRL method that uses voltage-sensitivity gradients to derive optimal VVO actions. A model-free multi-agent DQN framework for VVO and loss minimization was presented in [14]. In contrast, Xu et al. [15] employed radial basis function approximations to determine optimal transformer tap settings incorporating batch reinforcement learning. Furthermore, to improve scalability, several works have segregated ADNs into zone-level control areas [16], [17], [18], [19]. Cao et al. [16] introduced a surrogate ADN model supporting MA-DRL-based active power curtailment and PV reactive power scheduling for enhanced voltage regulation. A graph-based MA-DRL framework in [17] enabled coordinated decentralized VVO among zone agents. In [18], a global graph-attention surrogate model was integrated with MA-DRL to mitigate voltage deviations caused by anomalous measurements. Xiong et al. [19] developed a coordinated VVO approach that optimizes PV reactive power and adaptively modifies voltage-droop intercepts to reduce voltage deviations and losses. Furthermore, Zhang et al. [20] proposed an attention-based MA-DRL system for coordinating hybrid devices in renewable-rich ADNs, while Liu et al. [21] introduced a dual-critic DRL strategy to balance conflicting objectives by jointly training two critic networks and avoiding dominance of one objective over another.

Zhang et al. [22] introduced a two-timescale DDPG-based framework for coordinating VVO devices, including CBs, VRs, and ESSs. The study in [23] presented a two-stage DRL approach for the coordinated voltage control of OLTCs, CBs, and EVs. A multiobjective DRL strategy is presented in [24] for voltage regulation and unbalance mitigation in LVDNs through optimal control of distributed generation. Furthermore, Wang et al. [25] presented a multi-agent DRL framework that coordinates EVs and PV systems via active and reactive power control to regulate voltage in active distribution networks.

Existing centralized training and decentralized execution (CTDE) frameworks for distribution networks rely on area-level centralized training that uses concatenated states and actions across all agents to ensure stationarity in multi-agent actor–critic environments. In low-voltage distribution networks (LVDNs), this leads to significant communication, memory, and computing demands as the dimension of the critic input increases with network size, making node-level optimization difficult. Furthermore, implementing a shared global reward pushes all agents, including those without local voltage violations, to respond to system-wide events, thereby diminishing convergence efficiency. The resilience of these frameworks during communication disruptions is frequently unaddressed, despite their significant reliance on uninterrupted global information exchange. Table I provides a comparative analysis of existing methodologies and the proposed framework.

C. Contributions

The proposed framework addresses the limitations of conventional CTDE in a multi-agent environment by decomposing VVO/VWO into two complementary learning tasks: i) centralized network loss minimization and ii) decentralized node voltage control. Unlike CTDE methods that propose a common critic and a uniform reward framework during fully centralized training, the proposed centralized–decentralized training and execution (CDT&E) architecture distinctly separates global and local objectives. Decentralized agents independently execute priority-based voltage control, while system-level loss minimization is achieved by a modified centralized training module that employs a weighted multiobjective (multicritic) framework. A stagewise control technique is implemented, in which VVO is primarily executed in stage 1.1, followed by VWO in stage 1.2, only when required to prevent excessive active power curtailment. This work's primary contributions are summarized as follows.

1) In the decentralized objective, each agent regulates voltage solely based on local observations and actions, thereby avoiding the state–action aggregation inherent in CTDE frameworks. Computation is entirely decentralized, thereby, minimizing communication costs, enhancing scalability, and preventing single-point failure. A reward-backstepping-based contribution-factor approach is presented to address non-stationarity in independent multi-agent learning, thereby facilitating precise credit assignment and stable policy convergence.
2) Centralized training is crucial for minimizing losses, as it facilitates coordinated decision-making by leveraging global information. An action transformation scheme

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TABLE I
TAXONOMY OF THE EXISTING AND PROPOSED FRAMEWORK

Modeling Type	Ref.	Control Architecture	Communication Dependency	Convergence Stability	Generalization Ability	Scalability	Computational Efficiency
Physics-based	[4]	Centralized	High	High	Low–Medium	Low	Low
	[5]	Distributed	Medium–High	High	Medium	High	Medium
	[6]	Decentralized	Low	Very High	Low	Very High	Very High
	[8]	Distributed	Medium	High	Low–Medium	High	High
Hybrid (physics + data)	[7]	Decentralized	Low–Medium	Medium	Medium	Medium	Medium
Hybrid-based DRL	[18]	Distributed	Medium	High	High	High	Medium
	[20]	Distributed	Medium–High	High	Low	Medium	Low
	[22]	Distributed	Medium–High	High	Medium	Medium	Medium
Data-driven interactive DRL	[11]	CTDE	Medium	Medium	Medium	High	Low
	[12]	CTDE	Medium–High	Medium	High	Medium	Low
	[13]	Distributed	Medium	Medium	Medium	High	Low
	[16]	Decentralized	Low–Medium	Medium	Medium	High	Low
	[17]	Distributed	Medium	High	High	High	Medium
	[21]	CTDE	Medium	High	Medium	High	Low
	[23]	Distributed	Medium–High	Medium	Medium	High	Low
	[24]	Distributed	Medium–High	Medium	High	Medium	Low
	[25]	Distributed	High	Medium	Medium	High	Low
	Proposed	Weighted centralized-decentralized	Low–Medium	High	High	Very High	High

is proposed to enhance scalability by compressing the joint action space and integrating it with low-dimensional shared global states. This approach reduces the input dimension for the centralized critic, thereby enhancing computational efficiency in large-scale DRL optimization problems.

3) A position-timing (POST) sequence is developed to maintain voltage regulation during partial or complete loss of communication with the central server. The POST mechanism leverages the radial structure and temporal organization of general distribution networks to facilitate pseudocoordination among agents. This approach ensures sequential operational adjustments to maintain local voltage regulation.

The efficacy and scalability of the proposed optimization framework are assessed on a practical distribution network, along with the modified IEEE 33 and IEEE 119-bus system, under various operating conditions. Section II presents the mathematical formulation of the proposed VVO/VWO problem and the corresponding DRL algorithm components. The proposed framework implementation and discussion is detailed in Section III, while the results and comparative performance evaluation are provided in Section IV.

II. PROPOSED VOLT-VAR AND VOLT-WATT OPTIMIZATION PROBLEM FORMULATION

Fig. 1 illustrates the two categories of consumers in the ADN under study: 1) single distribution transformer (DTR) single consumers (SDSC) and 2) single DTR multiple consumers (SDMC). Each consumer in SDSC is provided with a dedicated DTR, IB-DER, and pump load. In SDMC, several consumers are connected to a single DTR, each with its own IB-DER and pump load. Each SDSC and sector under each SDMC DTR is referred to as an area of control (AoC) to segregate the distribution network into smaller zones. The objective function of the hierarchical VVO/VWO problem for any AoC is summarized in (1)–(9).

$$Stage\ 1.1: OF_{1.1} = \min_{\mathbf{a_t}} \left[P_t^l\right] \tag{1}$$

$$s.t.\ \ \underline{V} \le V_{k,t} \le \overline{V};\ \ \forall\ i \in N_b, t \in [0,T) \tag{2}$$

$$\sum_{k=1}^{n_c} P_{k,t}^d + P_t^l = P_t^{grid} + \sum_{k=1}^{n_c} P_{k,t}^g \tag{3}$$

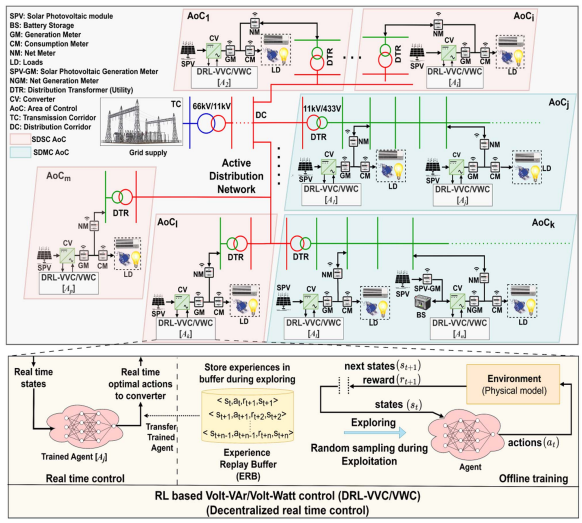


Fig. 1. Representative structure of complete control framework.

$$\sum_{k=1}^{n_c} \mathbf{Q}_{k,t}^d + \mathbf{Q}_t^l = \mathbf{Q}_t^{grid} + \sum_{k=1}^{n_c} \mathbf{Q}_{k,t}^g \tag{4}$$

$$\mathbf{Q}_{k,t}^g = \mathbf{a_{k,t}} P_{k,t}^g;\ \ -\tan(\tau_{th}) \le \mathbf{a_{k,t}} \le \tan(\tau_{th}) \tag{5}$$

$$S_{k,t} = \sum_{j=1}^{n_c} V_{k,t} V_{j,t} Y_{kj} \angle(\delta_{j,t} - \delta_{k,t} + \Theta_{k,j}) \tag{6}$$

$$Stage\ 1.2: OF_{1.2} = \min_{\mathbf{b_t}} \left[P_t^l + \sum_{k=0}^{n_c} (P_{k,t}^{g_0} - P_{k,t}^g)^2\right] \tag{7}$$

$$s.t.\ \ (2)-(6);\ P_{k,t}^g = \mathbf{b_{k,t}} P_{k,t}^{g_0};\ b_{th} \le \mathbf{b_{k,t}} \le 1 \tag{8}$$

$$\mathbf{Q}_{k,t}^g = \pm\tan(\tau_{\text{th}}) P_{k,t}^g;\ (P_{k,t}^g)^2 + (\mathbf{Q}_{k,t}^g)^2 \le (S_k^g)^2 \tag{9}$$

where, τ_{th} is the power factor angle threshold, $\mathbf{a_t} = [a_{1,t}, a_{2,t}, \ldots, a_{n_c,t}]$, $\mathbf{b_t} = [b_{1,t}, b_{2,t}, \ldots, b_{n_c,t}]$ and "b_{th}" is the lower threshold factor of active power curtailment. The mathematical formulation of the optimization problem is given above, however, to realize the above problem in DRL form, the components required to model is discussed.

1) Episode: This article considers a one step MDP/MDG during training phase. Therefore, the length of each episode is one and the episode terminates when the agent/agents take an action both during the exploration and exploitation phase.

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2) State space: A decentralized partially observable state representation approach is considered in this study. Each SDMC area, is treated as a subdistribution network and each consumer connected to the LT network after the DTR is treated as an individual agent. Each agent in SDMC operate by observing their own local states and global states while, individually learning policies to achieve the individual and global objectives. The state vector of each consumer "k" at time "t" is defined as $\mathbf{s_{k,t}} = [V_t^{s,A_i}, V_{k,t}, P_{k,t}^d, \mathbf{Q}_{k,t}^d, P_{k,t}^g, \mathbf{Q}_{k,t}^g, n_t^{vl}, P_t^{l,A_i}]$, where, $V_{k,t}, P_{k,t}^d, \mathbf{Q}_{k,t}^d, P_{k,t}^g, \mathbf{Q}_{k,t}^g$ are the local states and $V_t^{s,A_i}, n_t^{vl}, P_t^{l,A_i}$ are the global states, n_t^{vl} is the number of violating nodes in an AoC. Similarly, the SDSC consumers also consider the same inputs which are inherently local.

3) Action space: Depending on the environment's conditions, each agent's actions are organized in a hierarchy. Initially, the action space related to the VVO problem is activated, and the control settings of the IB-DERs are altered to change the absorption/generation of reactive power. Further, the VWO module is activated only when the voltage violation is not mitigated in Stage 1.1 (DRL-VVO). Consequently, the IB-DER's active power generation is modified solely to attain the objectives with minimal curtailment of active power.

4) Reward function: The reward function is formulated considering the minimization of individual voltage violation and the global network loss of each AoC "A_i." Therefore, in this study, two reward functions denoting the AoC network loss minimization and individual node voltage regulation for all the stages are formulated as given below in (10)–(12), respectively.

$$R_t^{l,A_i} = -\left[w_1\left(\sum_{u=1}^{N_{TL}} P_{u,t}^l + P_{cu,t}^{l,D} + P_{nl,t}^{l,D}\right) + \frac{w_2}{N_b}\sum_{j=1}^{N_b}(vio_{j,t})\right] \tag{10}$$

$$r_{k,t}^{VVO,A_i} = -\left[\max\{(V_{k,t}-\overline{V}),0\} + \max\{(\underline{V}-V_{k,t}),0\} + \max\{(|a_{k,t}|-|a_{th}|),0\}\right] \tag{11}$$

$$r_{k,t}^{VWO,A_i} = -\left[\max\{(V_{k,t}-\overline{V}),0\} + \max\{(\underline{V}-V_{k,t}),0\} + \max\{(b_{th}-b_{k,t}),(b_{k,t}-1),0\} + (b_{k,t}P_{k,t}^{g_0} - P_{k,t}^{g_0})^2\right]. \tag{12}$$

Equation (10) defines the centralized AoC network-loss reward, incorporating line losses, DTR copper and no-load losses, and the number of voltage-violating nodes, and is applied uniformly across all stages. Equations (11) and (12) formulate the decentralized rewards for the VVO and VWO stages, respectively, based on nodewise voltage violations and stage-specific thresholds. These rewards quantify deviations from allowable voltage limits while respecting inverter technical constraints through a continuous action space. In the VWO stage, active power curtailment minimization (stage 1.2 objective) is explicitly embedded in (12), where "b_{th}" denotes the lower bound on curtailment. Conventional action clipping caused threshold saturation and slow convergence; therefore, penalty-based rewards without action clipping were adopted to keep actions within thresholds and accelerate convergence.

III. Centralized–Decentralized DRL Framework

This study integrates MADDPG with the proposed CDT&E method to improve optimality while reducing the number of training iterations and approximation error. MADDPG employs an actor–critic structure, in which the actor π_θ produces actions $a_t = \pi_\theta(s_t)$, and the critic assesses them through the state–action value function $Q_\phi(s_t, \pi_\theta(s_t))$. The critic parameters are revised utilizing the Bellman equation within the TD(1) framework, as in (13)-(15) given below:

$$o_t = \mathbb{E}\left[R_{t+1} + \gamma Q_{\phi'}(s_{t+1}, a_{t+1})\right]; a_{t+1} \sim \pi_{\theta'}(s_t) \tag{13}$$

$$\min_\phi L = \min_\phi (o_t - Q_\phi(s_t, a_t))^2; a_t \sim \pi_\theta(s_t) \tag{14}$$

$$\phi = \arg\min_\phi L. \tag{15}$$

After the evaluation step, the policy is updated by maximizing the Q-value during training the actor network and generate optimal action for the given state. This is achieved by executing gradient ascent to the loss function as in (16)-(18) given below:

$$\max_\theta J = \max_\theta (Q(s_t, \pi_\theta(s_t)) \tag{16}$$

$$\nabla_\theta J = \nabla_\theta \pi_\theta(s_t) \nabla_{\pi_\theta} Q(s_t, \pi_\theta(s_t)) \tag{17}$$

$$\theta_k = \arg\max_\theta J. \tag{18}$$

A. Proposed VVO-VWO Framework for SDMC/SDSC

The proposed multi-agent DRL framework provides agents to manage local voltages independently while minimizing network losses collaboratively. Therefore, two critics train on conflicting objectives; one is trained through independent learning, and the other through coordinated learning. Independent learning is susceptible to misleading rewards influenced by external interactions, leading to incorrect credit assignment and non-stationarity. Therefore, to address this challenge, a reward backstepping-based algorithm for determining the contribution factor is introduced, which is discussed below.

1) Reward backstepping-based contribution factor: To address the above challenge, a *reward backstepping-based contribution factor* "$\zeta_{k,t}$," defined in (19) is proposed. In existing CTDE methods, stationarity was maintained through complete observability of local states to each agent. However, the proposed formulation avoids aggregating joint states and actions, particularly for objective 1 (decentralized objective). Therefore, *contribution factor* becomes an alternative tool to manage non-stationarity during independent learning.

$$\zeta_{k,t} = -\left(\frac{r_{k,t}^v(s_{k,t}, a'_{k,t}, s''_{k,t})}{r_{k,t_0}^v(s_{k,t}, a_{k,t}, s'_{k,t}) - \epsilon}\right);\ R_{k,t}^{v,A_i} = \zeta_{k,t};$$
$$\epsilon > 0;\ \ v \in \{VVO, VWO\} \tag{19}$$

The computation involves initially assessing the local reward $r_{k,t_0}^v(s_{k,t}, a_{k,t}, s'_{k,t})$ of agent "k" based on the collective initial actions of all agents, followed by a re-evaluation that modifies only the target agent's action either during exploration or policy exploitation while keeping the initial policies of the other agents fixed to determine the modified reward $r_{k,t}^v(s_{k,t}, a'_{k,t}, s''_{k,t})$ of agent "k" as shown in Fig. 2.

The contribution factor $\zeta_{k,t}$ therefore serves as a measure of the agent's marginal impact on its local environment. Although

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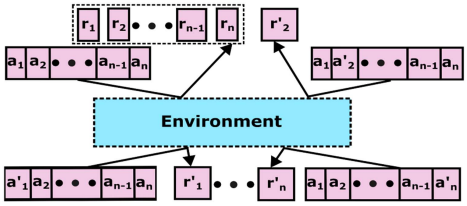


Fig. 2. Reward backstepping-based contribution factor design.

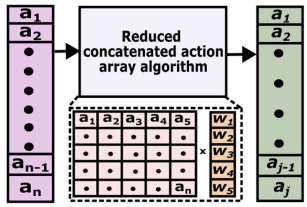


Fig. 3. Block diagram of dimensionality reduction algorithm.

full stationarity is not restored, it is effectively managed by isolating policy updates to a single agent while holding the remaining agents fixed. This allows each agent to experience state transitions driven by its own actions under a quasi-stationary external environment. The procedure is repeated for all agents, after which the joint action is updated only if the new policy improves $\zeta_{k,t}$; otherwise, the previous action is retained. If an agent's action improves the contribution factor, i.e., $r^v_{k,t}(s_{k,t}, a'_{k,t}, s''_{k,t}) < r^v_{k,t_0}(s_{k,t}, a_{k,t}, s'_{k,t})$, the policy update progresses toward a favorable region of the search space, improving local rewards and Q-values and facilitating convergence of both the critic and actor gradients, $\nabla_\theta J_k \rightarrow$ small. Conversely, a deterioration in $\zeta_{k,t}$, i.e., $r^v_{k,t}(s_{k,t}, a'_{k,t}, s''_{k,t}) > r^v_{k,t_0}(s_{k,t}, a_{k,t}, s'_{k,t})$, signals sub-optimal/deteriorating policy updates and prompts corrective adaptation. A brief discussion on the convergence analysis and credit assignment via contribution factor ($\zeta_{k,t}$) is provided in (20)–(24) below. Further, a detailed discussion on the convergence analysis and the credit assignment technique of this design is given in appendix A.1 and appendix A.2 respectively of the supplementary file.

a) Convergence analysis via contribution factor: To ensure stable learning behavior, we analyze the contribution factor $\zeta_{k,t}$ defined in (20) where rewards are assumed negative, i.e., $r^v_{k,t}, r^v_{k,t_0} \leq 0$. The constant "$\epsilon$" is selected such that $0 < \epsilon < 1$ to prevent denominator singularity and preserve the sign of the ratio. Intuitively, when $\zeta_{k,t} \rightarrow 0$, the current actions yield improved rewards compared to the initial baseline r^v_{k,t_0}, indicating correct learning progression. Conversely, $\zeta_{k,t}$ moving away from zero implies policy degradation. To perform the analysis, a Lyapunov function is defined as in (20).

$$V(t) = \frac{1}{2}\sum_{k=1}^{N} \zeta^2_{k,t} \tag{20}$$

where $V(t)$ measures the aggregate deviation from the desired equilibrium $\zeta_{k,t} = 0$. If actor updates are oriented such that the expected change in $\zeta_{k,t}$ satisfies as given in (21).

$$\mathbb{E}[\Delta\zeta_{k,t} \mid \mathcal{P}_t]\,\zeta_{k,t} \leq -\rho_k \zeta^2_{k,t}; \qquad \rho_k > 0 \tag{21}$$

then the expected Lyapunov difference satisfies as in (22).

$$\mathbb{E}[V(t+1) - V(t) \mid \mathcal{P}_t] \leq -\rho V(t) + C\alpha^2 \tag{22}$$

where "α" is the step-size, "C" is a bounded constant, and "ρ" is the Lyapunov decay rate. Hence, for sufficiently small "α," the multi-agent learning process is locally stable and "$V(t)$" decreases in expectation, implying convergence of "$\zeta_{k,t} \rightarrow 0$." Here, "$\mathcal{P}_t$" represents all information available up to time "t."

b) Discussion on contribution-factor based credit assignment: Consider a multi-agent system with agents $k \in \{1, \ldots, N\}$ each agent having a local reward $r^v_k(s, a)$ that is continuously differentiable in the joint action $a = (a_1, \ldots, a_N)$. Let the contribution factor computed at time "t" by a small perturbation of agent k's action be represented as in (23):

$$\zeta_{k,t} \;=\; -\left(\frac{r^v_{k,t}\,(s_t,\, a_{k,t} + \Delta a_k,\, a_{-k,t})}{r^v_{k,t}\,(s_t,\, a_{k,t},\, a_{-k,t}) - \epsilon}\right) \tag{23}$$

where $a_{-k,t}$ denotes other agents' actions fixed during the perturbation considering Δa_k is small. Let the policy of agent "k" be $\mu_{\theta_k}(s)$ then the policy gradient is defined as in (24).

$$g_{\theta_k}(s) = J_{\mu_k}(s)^\top \nabla_{a_k} r^v_k(s, a); \quad J_{\mu_k}(s) = \frac{\partial \mu_{\theta_k}(s)}{\partial \theta_k} \tag{24}$$

Under Assumptions (A1)–(A4) given below, the deterministic policy-gradient update computed using a decentralized critic trained toward "$\zeta_{k,t}$" is aligned (in expectation) with "g_{θ_k}" up to a positive scalar factor and bounded error terms. Consequently, using "$\zeta_{k,t}$" for critic/actor updates yields correct (directional) credit assignment for agent "k."

- A1) *Smoothness:* $r^v_k(s, a)$ and $\mu_{\theta_k}(s)$ are continuously differentiable; gradients and Jacobians are locally bounded.
- A2) *Small perturbation:* $\|\Delta a_k\|$ is sufficiently small so that first-order Taylor approximations are accurate.
- A3) *Timescale separation:* When measuring the perturbed and baseline rewards used to compute $\zeta_{k,t}$, other agents' actions $a_{-k,t}$ are effectively constant.
- A4) *Critic regression:* The decentralized critic $Q^d_{\phi_k}(s_{k,t}, a_{k,t})$ is trained so that, in expectation, $Q^d_{\phi_k}(s_{k,t}, a_{k,t}) \approx \mathbb{E}[\zeta_{k,t} \mid s_{k,t}, a_{k,t}]$.

2) Dimensionality reduction (DR) of concatenated action array: This study also incorporates centralized training to achieve global goal of network loss minimization, which needs direct agent co-ordination. The critic network receives the concatenated local states and actions $Q^{[\pi]}_{\phi'_{l,t}}([s]_t, [a]_t)$ from each agent in the traditional centralized multi-agent DRL training strategies. However, as the number of agents rises, this technique may require extensive memory, processing power to handle parameters (specially input layer parameters) of the critic network leading to inflated parameters and slower algorithm execution during training. Therefore, to overcome this problem, this article offers an approximation technique that reduces action array length and assures a nonidentical representation for any concatenated action array without affecting the differential chain of layers during backpropagation. Fig. 3 depicts the block diagram of action dimensionality reduction algorithm while describing the transformation strategy in Algorithm 1. The reduced action array is concatenated with only the global states $[V^{s,A_i}_t, n^{vl}_t, P^{l,A_i}_t]$, which are then fed as inputs to the designated critic and actor network. An extended discussion on this algorithm is provided in Appendix B of the supplementary file.

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Algorithm 1: Reduced concatenated action array.

Input: Concatenated action of each agent $[a]_t$
$n_{rows} \leftarrow round(ln(length([a]_t)))$
$n_{cols} \leftarrow \frac{length([a]_t)}{n_{rows}}$
While $n_{cols} \; != I^+$ **do**
$zero_pad([a]_t)$
end while
$[a_{res}]_t \leftarrow$ Reshape$([a]_t$; shape $\leftarrow [n_{rows}, n_{cols}])$
Initialize a trainable weight vector $[W]$ of size $(n_{cols} \times 1)$
$[a_{tns}]_{n_{rows} \times 1} \leftarrow matmul([a_{res}]_{n_{rows} \times n_{cols}}, [W]_{n_{cols} \times 1})$
Output: Reduced concatenated action array $[a_{tns}]$

B. Centralized-Decentralized Algorithm

The training framework in (25)–(29) present the VVO/VWO centralized–decentralized training structure:

$$Q^{\pi_k}_{\phi'_{v,k,t}}(s_{k,t}, a_{k,t}) = \mathbb{E}_{a_k \sim \pi_k}\left[R^{v,A_i}_{k,t} | s_{k,t}, a_{k,t}\right]$$

target critic for voltage violation with $\gamma = 0$ (25)

$$Q^{[\pi]}_{\phi'_{l,t}}([s_g]_t, [a_{tns}]_t) = \mathbb{E}_{[a_{tns}]_t \sim [\pi]}\left[R^{l,A_i}_t | [s_g]_t, [a_{tns}]_t\right]$$

target critic for network loss minimization with $\gamma = 0$ (26)

where (25) and (26) represent the target critic networks parameterized by ϕ'_v and ϕ'_l for an one step MDG. It is to be noted that "$x_{k,t}$" denotes the respective variable for a single consumer/agent while "$[x]_t$" represents the aggregated variable. The voltage violation mitigation is defined as a local objective (obj. 1), thus, the inputs to the critic networks $Q_{\phi'_{v,k,t}}$ and $Q_{\phi_{v,k,t}}$ are observations of local states $s_{k,t}$, actions $a_{k,t}$ and reward $R^{v,A_i}_{k,t}$ for a consumer "k" at time "t." However, for network loss minimization the objective is global (obj. 2) and requires the global state of the AoC therefore the global information state space $[s_g]_t$, reduced joint action space $[a_{tns}]_t$ and the global reward R^{l,A_i}_t are fed as inputs to the critic networks $Q_{\phi'_{l,t}}$ and $Q_{\phi_{l,t}}$ to establish co-ordination among the agents as shown in (25) and (26). The critic networks $Q_{\phi_{v,k,t}}$ (decentralized critic) and $Q_{\phi_{l,t}}$ (centralized critic) are trained by minimizing the following loss equations in (27) and (28) given below.

$$L_{\phi_{v,k,t}} = \frac{1}{|e_k|}\sum_{e_k \in B^d_k}\left[\left(Q_{\phi_{v,k,t}}(s_{k,t}, a_{k,t}) - Q_{\phi'_{v,k,t}}(s_{k,t}, a_{k,t})\right)^2\right] \quad (27)$$

$$L_{\phi_{l,t}} = \frac{1}{|e|}\sum_{e \in B^c}\left[\left(Q_{\phi_{l,t}}([s_g]_t, [a_{tns}]_t) - Q_{\phi'_{l,t}}([s_g]_t, [a_{tns}]_t)\right)^2\right]$$

$e_k \in [s_{k,t}, a_{k,t}, R^{v,A_i}_{k,t}], e \in [[s_g]_t, [a_{tns}]_t, R^{l,A_i}_t]$ are the sampled experiences for (27) and (28), respectively. (28)

Further, to train the actor network of agent "k" the loss function is given as in (29):

$$L_{\theta_{k,t}} = \frac{1}{|e'|}\sum_{e' \in B}\Big[w'\, Q_{\phi_{v,k,t}}(s_{k,t}, a_{k,t}) + (1-w')\left(Q_{\phi_{l,t}}([s_g]_t, [a_{tns}]_t)\right)^2\Big]; \quad B \in B^d_k \cup B^c \quad (29)$$

where B^d_k & B^c are the replay buffers corresponding to decentralized and centralized critic, respectively. The complete training strategy of the proposed framework is depicted in Fig. 4 while Fig. 5 illustrates the agent training and execution stages for voltage regulation with complete training algorithm given in Algorithm 2 of Appendix A in supplementary file.

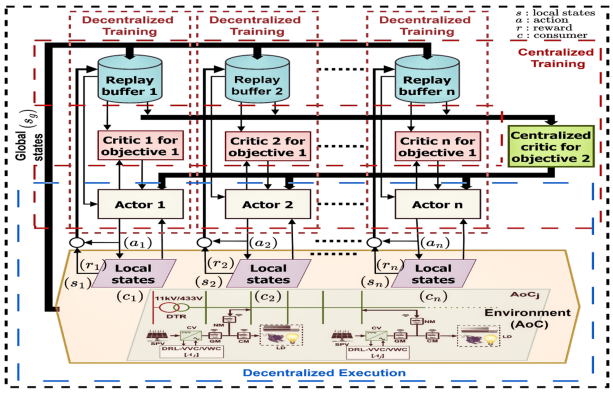


Fig. 4. Training strategy of proposed CDT&E framework.

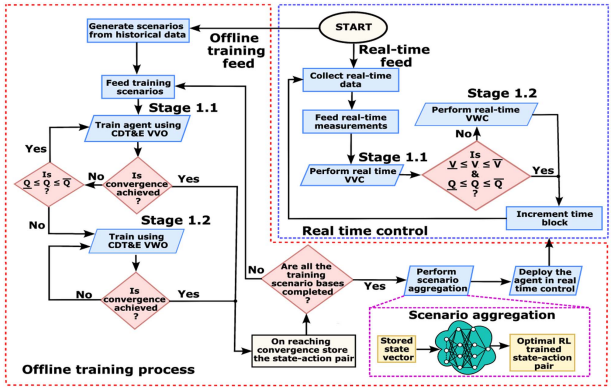


Fig. 5. Flowchart of the proposed hierarchical control framework.

C. Position-Timing (POST) Sequence

This study's centralized objective necessitates the exchange of information across the network among agents, therefore requiring a strong communication support. However, the dependence on communication infrastructure poses serious concerns on the algorithm's reliability especially during partial or complete communication failure. To address this issue, a position-timing (POST) scheme is developed by leveraging the radial structure of the ADN to achieve pseudo-synchronization. During communication outages, agents adhere to a position-based update sequence, which involves splitting the network into "y" positional layers as shown in Fig. 6. Agents at the "$y_{i_{th}}$" position revise their actions at time "t+i dt" based solely on local information. The operator decides the interval "dt" based on historical records of time taken by the network to reach steady state. Action updates propagate in sequence from the tail nodes to the root, maintaining network stability, with only those agents that encounter voltage violations and adjusting their operating points accordingly.

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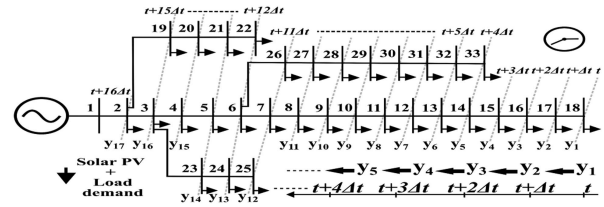


Fig. 6. Position-timing sequence in modified IEEE 33 bus system.

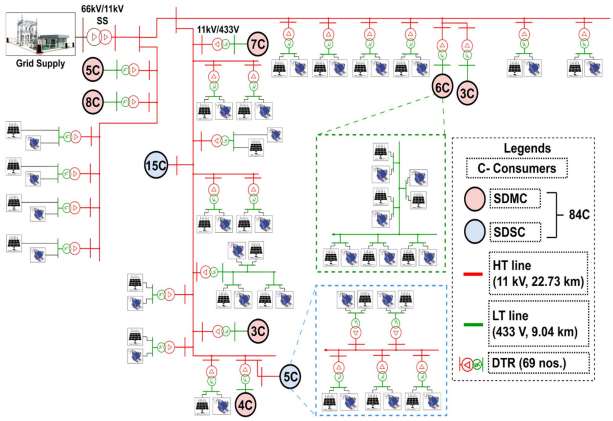


Fig. 7. Schematic diagram of the considered practical test system.

IV. Results and Discussions

A. Practical System Under Study

This study investigates a practical ADN shown in Fig. 7 supplying agricultural consumers with an overall contracted load of 693 HP and an installed solar capacity of 839 kWp, as part of the Surya-Shakti Kisan Yojna (SKY), a Government of Gujarat initiative for solarizing agricultural feeders. The dataset was obtained from the SKY Solar Energy Data Management (SEDM) platform of Gujarat Urja Vikas Nigam Limited (GUVNL) [26]. The network was modeled in DIgSILENT PowerFactory to produce balanced time-series load-flow solutions, including only grid-connected PV inverters as controllable devices, avoiding energy storage and tap-changing transformers. Loads and inverters were represented as nodal power injections/absorptions, with each consumer's PV rating (P_r^g) set at 1.25 times its rated load capacity (P_r^d). The DRL framework was implemented in TensorFlow (Python) and integrated with PowerFactory for closed-loop agent-environment interaction.

B. Simulation Results

The performance of the proposed hierarchical VVO/VWO framework is validated considering several cases for the execution of hierarchical stages, scalability analysis to validate its efficacy in larger networks and robustness validation during communication failures with parameter specifications given in Table II. The actor-critic network architectures were chosen empirically to optimize performance and computational efficiency. The actor's learning rate was set slightly lower than the critic to ensure stable policy updates and faster convergence, using Adam for its adaptable learning rates, momentum, and minimal tuning requirements. Initially, high-variance Gaussian noise was applied for exploration and progressively reduced during exploitation to facilitate deterministic action selection. The normalization weight (w'), which balances centralized and decentralized objectives, was determined through iterative simulations, as described in Appendix C of the supplementary file. Here, the various stages of VVO and VWO are discussed along with the condition for hard switching from VVO to VWO stages. Further, a performance comparison is given to compare the existing DRL frameworks with the proposed strategy.

TABLE II
Parameter Specifications of the Proposed Framework

Parameters	Network		Value
Number of hidden layers	Actor		2
	Cent./ Decent. Critic	State n/w	2
		Action n/w	1
		Concat.	2
Neurons in hidden layers	Actor		256, 128
	Cent./ Decent. Critic	State n/w	16, 32
		Action n/w	32
		Concat.	1024, 1024
Learning rate	Actor		7×10^{-4}
	Critic		1×10^{-3}
Optimizer	–		Adam
Activation functions	–		ReLU, Linear
Batch size	–		32
Replay buffer size	–		6×10^4
Exploration noise	–		$\mathcal{N}(0, 0.35)$
Exploitation noise	–		$\mathcal{N}(0, 0.0015)$
Exploration iterations	–		32
Thresholds	Voltage		[1.05, 0.95]
	Stage 1.1, 1.2- a_{th}, b_{th}		[-0.75, 0.75], [0.35,1]
Normalizing weights	Cent. Reward (w_1, w_2)		0.75, 0.25
	Decent. critic (w')		0.65
Singularity regularizer	ϵ		0.1

Scenario 1: Stage 1.1(a)- Overvoltage mitigation in SDMC AoC: The proposed control framework's performance in an SDMC AoC can be investigated by evaluating co-ordination and iterations to converge. The effectiveness of the proposed control strategy was compared to the CTDE approach [21] in terms of co-ordination and episode count for optimal policy establishment. The AoC considered comprises four consumers. Figs. 8 and 9 demonstrate the effectiveness of the proposed framework and CTDE approach. Figs. 8(a), 9(a), and 8(b), 9(b) illustrate reward and action improvements with both training methods. Figs. 8(c) and 9(c) show node voltage variations during training, while Figs. 8(d) and 9(d) exhibit reduced network loss in both approaches. The results show that the suggested strategy achieves an optimal policy in fewer episodes than CTDE while retaining co-ordination. Therefore, independent learning for local objectives and centralized training for global goals can solve the multiobjective problem faster and more effectively than the two critic-CTDE strategy.

Scenario 1: Stage 1.1(b)—Overvoltage mitigation in SDSC AoC: The proposed framework was reduced to a single-agent control framework to determine the optimal policy for a single consumer in an SDSC AoC. SDSC consumers do not need coordination; hence, CDT&E has not been implemented. Fig. 10(a) shows how rewards improve with each episode during the learning process in a given situation. The agent initially took random actions to explore the environment. After acquiring sufficient experience, the agent estimates decisions based on the policy/actor network. Fig. 10(b) exhibits the agent's actions trend as it learns the policy for a given environmental condition. Since the objectives are conflicting, careful consideration is required to obtain an optimal policy. The agent's activity is optimum when it minimizes network loss and achieves zero voltage violation while maintaining IB-DER operating limitations.

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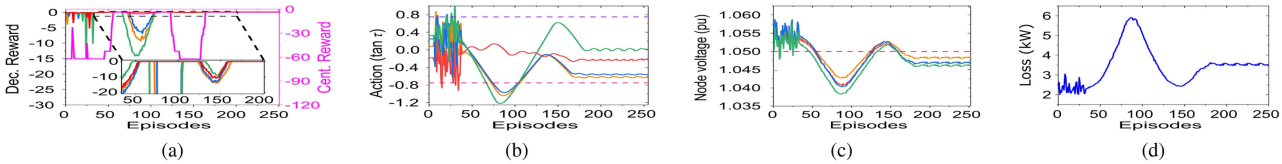


Fig. 8. Trend of (a) rewards, (b) actions, (c) node voltage, (d) network loss considering proposed CDT&E approach.

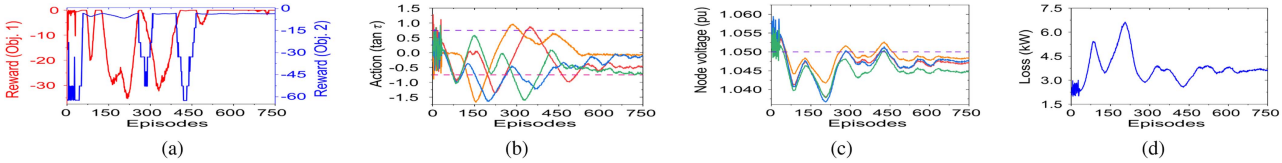


Fig. 9. Trend of (a) rewards, (b) actions, (c) node voltage, (d) network loss considering CTDE[21] approach.

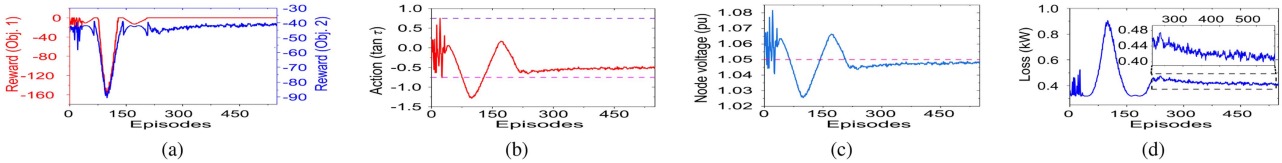


Fig. 10. Trend of (a) rewards, (b) actions, (c) node voltage, (d) network loss.

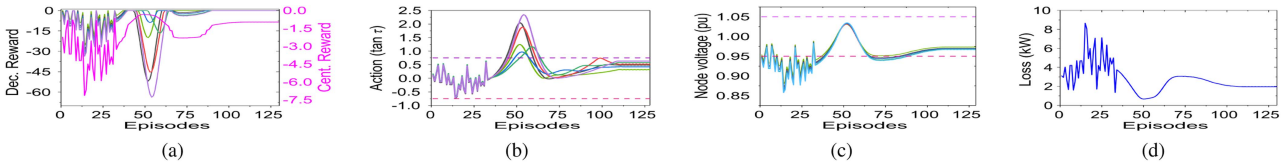


Fig. 11. Trend of (a) rewards, (b) actions, (c) node voltage, (d) network loss.

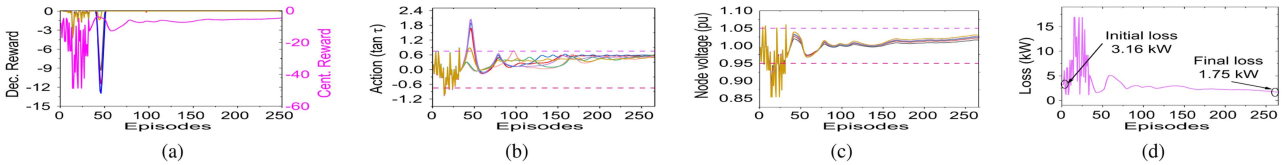


Fig. 12. Trend of (a) rewards, (b) actions, (c) node voltage, (d) network loss.

Figs. 10(c) and 10(d) show the node voltage is within the nominal band with corresponding changes in network loss during training.

Scenario 2: Stage 1.1—Undervoltage mitigation in SDMC AoC: The framework's performance was evaluated under several operating situations, including undervoltage scenarios. Initially, severe loading caused undervoltage, as depicted in Fig. 11. Fig. 11(a) illustrates the improvement of reward values, whereas Fig. 11(b) represents the trajectory of control actions taken to achieve the optimal operating values within the allowable range. The prevalence of positive actions signifies reactive power injection by IB-DER agents to facilitate voltage recovery. Fig. 11(c) verifies that node voltages were recovered within acceptable limits, while Fig. 11(d) demonstrate the improvement in AoC network losses. These results validate the proposed framework's effectiveness in regulating node voltage while minimizing network losses, even during undervoltage conditions without violating the technical constraints of IB-DERs.

Scenario 3: Stage 1.1—Loss minimization during no voltage violation: The framework was further assessed under operating conditions without voltage violations to evaluate its effectiveness in reducing network losses. The optimization in this scenario focused solely on reducing AoC loss, as illustrated in Fig. 12. Fig. 12(a) illustrates the improvement in reward values, whereas Fig. 12(b) depicts the corresponding control action trajectories that result in optimal voltage-compliant operation and achieving zero voltage violations finally. During the exploratory phase, transient voltage deviations were observed in Fig. 12(c), yet stable, nonviolating conditions were maintained throughout the training process. Fig. 12(d) shows that AoC losses got reduced from 3.16 to 1.75 kW. The results indicate that when decentralized objectives are inherently achieved, the framework

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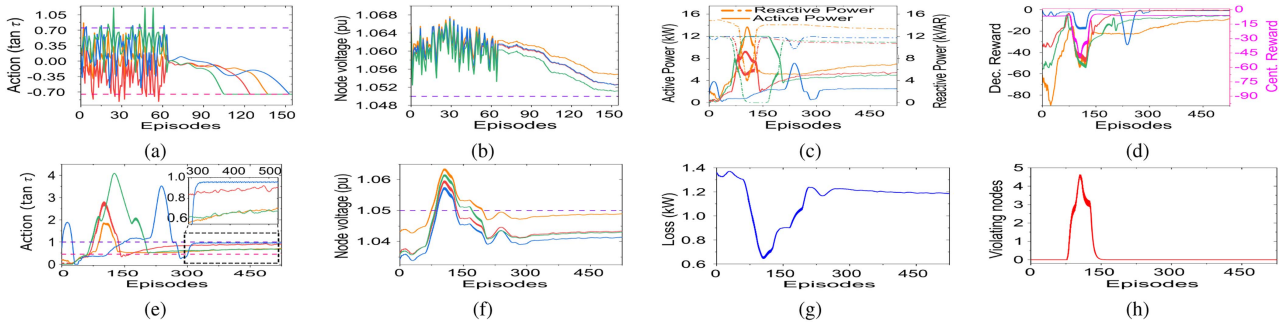


Fig. 13. Stage 1.1 (a) Actions, (b) voltage, Stage 1.2 (c) PV generation, (d) rewards, (e) actions, (f) voltage, (g) network loss, (h) violating nodes.

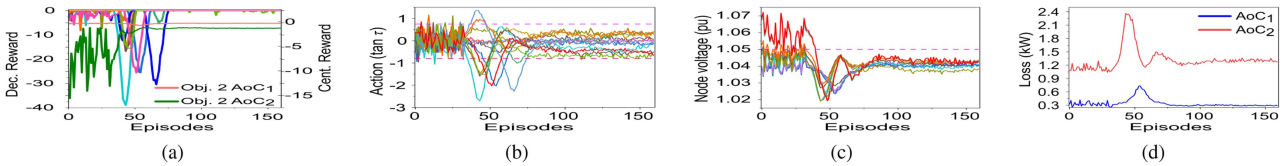


Fig. 14. Trend of (a) rewards, (b) actions, (c) node voltage, (d) power loss considering multiple AoCs of practical network.

successfully emphasizes the centralized loss-minimization goal by maintaining co-ordination among agents.

Scenario 4: Stage 1.2—Overvoltage mitigation with IB-DER constraints violation (VWO):

Under the operating constraints of the IB-DER, the capability for reactive power is represented as a function of active power generation from earlier stages. At this stage, the power factor constraint is relaxed to address the limitations encountered in stage 1.1, while maintaining the rated capacity limit of the IB-DER during the joint optimization of active and reactive power. Figs. 13(a) and 13(b) demonstrates that the stage 1.1 control structure is insufficient for addressing voltage violations within the allowable band, therefore, active power curtailment is essential for addressing voltage violations at this operating scenario. This situation requires the relaxation of power factor limits to facilitate effective co-optimization of active and reactive power, as illustrated in Fig. 13(c). Figs. 13(d) and 13(e) depicts the improvement of rewards and the change in actions taken by the agent, respectively. At first, the AoC node voltages appear to be within acceptable limits as seen in Fig. 13(f), but a closer examination in Fig. 13(e) reveals violations of active power constraints that is also undesired. However, the agents learn to recover the desired action space and at convergence, the framework significantly minimizes voltage violations and network losses while adhering to IB-DER operational constraints, as illustrated in Figs. 13(e)–13(h).

C. Scalability Validation of the Proposed Framework

The framework was first validated on multiple AoCs followed by modified IEEE 33 and IEEE 119 bus systems.

1) Validation considering multiple AoCs of practical network: The scalability evaluation of the proposed approach was executed by considering two SDMC AoCs with 5 and 8 consumers, respectively. Fig. 14 shows how the proposed approach minimizes voltage violations and network losses. Fig. 14(a) and (b) shows how agents in their AoCs establish an optimal strategy for their environment and improve their rewards and actions. Fig. 14(c) and (d) show how node voltage for each consumer and network loss vary in an AoC during learning. AoC_1 observed no violations, but AoC_2 initially experienced several violations. After the exploration phase, these violations were promptly minimized when the algorithm determined the optimal action set to reduce voltage violations while maintaining network loss and operational constraints.

2) Validation considering modified IEEE 33 bus system: The proposed method was evaluated on a modified IEEE 33-bus system, where loading levels and PV capacities were scaled using the SEDM dataset to generate diverse operating conditions exhibiting substantial voltage violations and network losses while considering the rating of solar PV (P_r^g) for each consumer is 1.25× the rated capacity (P_r^d) of the load (in kW) similar to the practical system conditions. Fig. 15 summarizes the framework's performance. Fig. 15(a) and (b) shows the agents' progressive learning of optimal strategies, reflected in steadily improving rewards. The corresponding node-voltage dynamics, final steady-state voltages, and network-loss profiles during training are presented in Fig. 15(c) and (d). Given the inherent conflict between decentralized voltage-regulation and centralized loss-minimization objectives, the framework prioritizes voltage-violation mitigation. Consequently, network loss increases during learning; however, the agents ultimately converge to an operating point that resolves voltage violations, network losses, while complying all constraints of stage 1.1.

3) Validation considering modified IEEE 119 bus system: The proposed method was further validated considering a IEEE 119 bus radial distribution system with modified loading and generating capacity for varying operating conditions. Similar to IEEE 33 bus system validation, the loads and PV generation were scaled from the SEDM dataset and the ratings of the devices are considered similar to the previous case. Fig. 16 depicts the algorithm's performance on this large network. Fig. 16(a) and (b) illustrates how agents/consumers learn an optimal strategy for the considered environmental condition to enhance their rewards (both local and global). Fig. 16(c) and (d) illustrates the node voltage fluctuations, consumer final voltage levels, and variation in network loss during training, respectively. Therefore, the algorithm attempts to mitigate violations by learning optimum actions and strike a balance between node voltage

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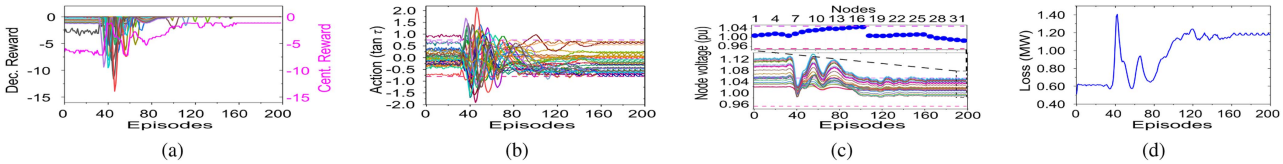


Fig. 15. Trend of (a) rewards, (b) actions, (c) node voltage, (d) network loss considering IEEE 33 bus system.

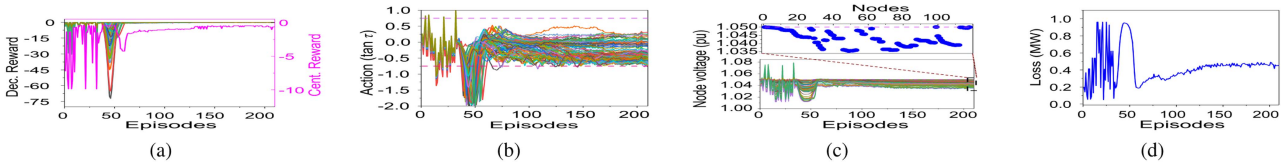


Fig. 16. Trend of (a) rewards, (b) actions, (c) node voltage, (d) network loss considering IEEE 119 bus system.

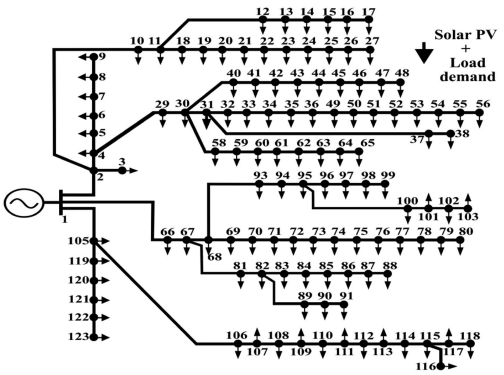


Fig. 17. Single line diagram of modified IEEE 119 bus system.

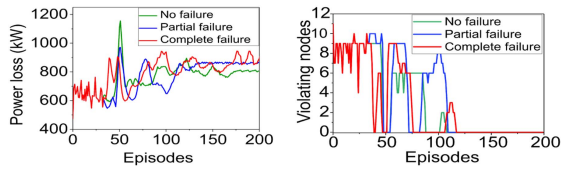


Fig. 18. Variation in (a) network losses, (b) voltage violation due to communication failures during training.

TABLE III
NETWORK LOSS DURING VARIOUS COMMUNICATION STATUS

Communication status		No failure	Partial failure	Complete failure
Network losses (MW)	Average (μ)	1.26	1.34	1.39
	Standard deviation (σ)	0.76	0.74	0.73

violation minimization and network loss minimization while complying with the operational constraints in modified IEEE 119 bus system given in Fig. 17.

D. Robustness Validation of the Proposed Framework

ADNs are vulnerable to communication failures, which can impact both the training and execution phases of multiagent controllers. The proposed framework addresses voltage regulation through decentralized learning, while minimizing network loss through centralized learning. In the events of communication failure, agents are unable to access global state information, including slack-bus voltage (V_t^{s,A_i}), network loss (P_t^{l,A_i}), and the number of violating nodes within an AoC (n_t^{vl}). However, they retain their local observations, which allows them to persist in regulating their own voltages. Three case studies were conducted on the modified IEEE 33-bus system to assess the framework's robustness under different communication conditions.

1) Validation during communication failures:

Base case- No communication failure: In this scenario, all agents have access to complete global states and rewards, ensuring effective communication. Fig. 18(a) illustrates that this condition produces the lowest network losses, reflecting efficient co-ordination.

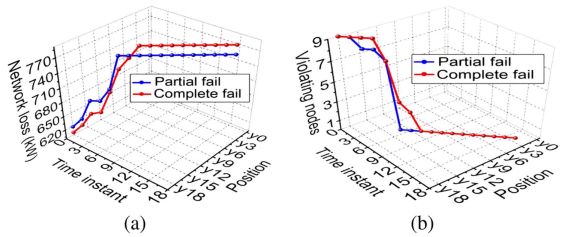
Case 1- Partial communication failure in AoC: In this case, buses 2, 8, 13, 18, and 22 lacked access to global information. Agents operating at these buses rely exclusively on local observations, focusing on local voltage correction while avoiding the system-wide impacts. As a result, Fig. 18(a) illustrates greater network losses than those in the base case, yet still lower than in the complete-failure scenario (case 3). However, agents that maintain effective communication links perform an essential role in reducing losses, demonstrating partial resilience toward loss minimization.

Case 2—Complete communication failure in AoC: In this scenario, all agents are deprived of access to global states and rewards. Only decentralized voltage regulation is currently in operation, resulting in the neglect of the centralized loss-minimization objective. Therefore, Fig. 18(a) depicts the highest network losses along with increased oscillatory behavior.

Nevertheless, since the decentralized objective remains active across all three cases, the framework continues to mitigate individual node voltage violations. This is evident from Fig. 18(b), where the number of nodes violating the desired voltage threshold ultimately converges to zero in all the three cases. Table III gives the average loss incurred in the network for various operating condition considering no failure, partial failure, and complete communication failure. It is again observed that the average loss is the least during no communication failure, followed by increase in network loss during partial failure with the maximum network losses incurred during complete failure.

2) POST sequence: The proposed POST sequence demonstrates significant effectiveness, especially in cases of partial or complete communication failure. Similar scenarios (case 2 and case 3) were performed for real-time validation to evaluate their robustness. The absence of centralized communication results

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(a) (b)

Fig. 19. (a) Network losses. (b) Voltage violation due to POST sequence.

TABLE IV
POST SEQUENCE RESULTS FOR AN OPERATING SCENARIO

Communication status	No failure	Partial failure	Complete failure
Network loss (MW)	0.73	0.76	0.78
Positions to zero violations (y_i)	-	5	8

in to network non-stationarity, which restricts agents' access to global state information and negatively impacts decision-making. The POST sequence addresses this challenge by facilitating pseudocoordination through the use of spatio-temporal relationships, specifically agent positions, the temporal ordering of actions, and the radial system topology. Fig. 19(a) indicates a progressive decrease in voltage violations, with conditions of zero violations attaining faster during partial failure compared to complete failure, aligning with the expected outcomes. Fig. 19(b) illustrates that steady-state network losses increase significantly during complete communication failure, as all agents are denied access to the central coordinator. In the event of a partial failure, the connected agents have the capacity to compensate for AoC losses by adjusting voltages and helping to minimize losses. Table IV presents the average losses under various test conditions, highlighting the robustness of the POST sequence in environments with communication constraints.

E. Performance Assessment

This section discusses on the performance of the proposed framework on various parameters:

1) Computational complexity analysis: The proposed technique incorporates a local exploratory evaluation to calculate the contribution factor "$\zeta_{k,t}$," which results in a slight increase in computational overhead. However, the overall computation involved in the training process is significantly reduced due to the disaggregation of agent states and actions during decentralized training. Table V compares the per-update computational cost of a traditional CTDE actor-critic-based methodology with the proposed contribution-factor ("ζ") framework. The computational benefits can be summarized as follows.

1) The proposed framework increases the per-episode environment interactions to N (from 1) due to auxiliary exploratory steps; however, parallel execution of distributed agents reduces the overall training time and computational burden.
2) The dominant CTDE complexity term, $\mathcal{O}(N^2 d_\theta)$, is reduced to $\mathcal{O}(N d_\theta)$ via contribution-factor decoupling and dimensionality reduction, yielding sublinear scaling of actor/critic updates.
3) Communication effort reduces from transmitting all joint actions "$\mathcal{O}(N d_a)$" to sending only compressed actions and scalars "$\mathcal{O}(d_{tns})$," which further improves wall-clock efficiency in distributed parallel processing.

TABLE V
COMPARISON OF COMPUTATION COMPLEXITY

Component	Quantity (per iteration)	CTDE framework	Proposed framework
Environment interactions	Environment transitions per iteration	N joint actions $\Rightarrow \mathcal{O}(1)$	1(baseline step) $+ n^{vl}N$ $\Rightarrow \mathcal{O}(1 + n^{vl}N)$ parallelized computation
Critic update update	Neural compute per critic update	$\mathcal{O}(d_\phi N(d_s + d_a))$ $+\mathcal{O}(N^2 d_\theta)$(coupling)	$\mathcal{O}(N d_\phi d_s + d_{\phi^c} d_{tns})$ $+\mathcal{O}(N d_\theta)$ (coupling)
Actor update cost	Policy gradient evaluation	$\mathcal{O}(N^2 d_\theta)$ Centralized update	$\mathcal{O}(N d_\theta)$ Decentralized + Reduced Centralized update
Communication overhead	Data transmitted per step	Joint actions $(a_1, .., a_N)$ of size $\mathcal{O}(N d_a)$	Compressed joint actions $[a_{tns}]$ of size $\mathcal{O}(d_{tns})$
Overall per-update		$\mathcal{O}(d_\phi N(d_s + d_a))$ $+\mathcal{O}(N^2 d_\theta)$	$\mathcal{O}(N d_\phi d_s + d_{\phi^c} d_{tns})$ $+\mathcal{O}(N d_\theta)$
Scalability	as $N \to \infty$	Quadratic $\mathcal{O}(N^2)$	Sublinear $\mathcal{O}(d_{tns} N)$

N = No. of agents, d_s = state dim. per agent, d_a = action dim. per agent, d_θ = actor parameter dim. per agent, d_ϕ = per-critic parameter dim., d_{ϕ^c} = centralized-critic parameter dim., d_{tns} = reduced joint-action dim. after DR.

TABLE VI
COMPARISON OF EXISTING AND PROPOSED FRAMEWORKS

Method	Bus System	Iterations		Final rewards (average)		% of successful converging scenarios (VVO)
		Average	Standard deviation	Obj. 1	Obj. 2	
Heuristic algorithm (PSO)	**Practical**	237	255	–		93.21
	33	438	370			85.35
	119	510	150			76.23
Conventional DRL (CTDE) [13]	**Practical**	300	38	−15.73		97.32
	33	285	62	−85.37		90.56
	119	400	88	−100.25		83.47
Two-critic DRL (CTDE) [21]	**Practical**	223	28	−9.24		98.23
	33	250	43	−75.21		95.23
	119	325	67	−91.05		91.21
Proposed	**Practical**	132	35	−0.35	−5.1	99.23
	33	185	60	−0.95	−36.6	98.71
	119	237	82	−1.75	−73.15	97.95

2) Performance comparison of the proposed framework with existing techniques: The performance comparison shown in Table VI demonstrates that the proposed framework outperforms existing approaches in most evaluation criteria. It can be observed that the proposed framework required the least iterations across all evaluated network topologies and achieved the highest final rewards among the compared strategies. The proposed strategy also achieved the highest percentage of successful convergence across 500 training scenarios, solving majority of the cases using VVO alone. Further, the framework was also evaluated against a heuristic PSO-based method where the maximum iterations (750) were kept as hard stopping criterion with an early stopping criterion of tolerance value tol ≤ 1000. It was observed that the PSO algorithm achieved convergence very quickly in some scenarios but demonstrated a significantly lower rate of convergence success since it hit the hard stopping criterion in most of the scenarios as compared to the DRL-based methods exhibiting saturation without reaching the desired convergence level, and thereby, increasing the average iterations as observed in Table VI. The proposed framework consistently delivered more optimal and reliably convergent solutions across various scenarios when compared to both heuristic and existing DRL methods.

3) Discussion on practical deployment of the proposed framework: The proposed framework addresses significant limitations of current DRL-based methods, including excessive computational demands, strict interagent reward dependencies, slow convergence rates, and substantial communication requirements, especially in large-scale multi-agent environments. The framework enables parallel computation and decentralized

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decision-making by assigning dedicated, distributed processors to individual agents, thereby reducing dependence on centralized processing. The effectiveness is initially demonstrated on a practical distribution feeder across various operating conditions (see Section IV-B). Further, scalability is verified through studies that examine multiple AoCs on the same feeder, as well as on modified IEEE 33-bus and IEEE 119-bus systems, demonstrating consistently satisfactory results (see Section IV-C). The evaluation of robustness is also conducted considering partial and complete communication failures (see Section IV-D) that is compared with the base case of no-communication failures. The framework has been validated in a simulated environment and can be easily adapted for real-world applications with minimal infrastructure improvements. However, currently, grid-integrated PV inverters primarily operate at a unity power factor and have limited communication capabilities, resulting in practical challenges during their deployment. Therefore, implementing the proposed framework requires specific enhancements, such as integrating low-cost, IoT-enabled microprocessors (like Raspberry Pi or comparable high-speed controllers) and adopting flexible reactive power control in both grid-connected and standalone PV inverters, following suitable regulatory guidelines.

V. Conclusion

This article presents a data-driven optimization and control framework that uses multi-agent deep reinforcement learning for Volt–VAr and Volt–Watt optimization in ADNs. A CDT&E architecture is developed to regulate nodal voltages and minimize network losses through pseudo co-ordinated control of PV inverter operating points. The framework decomposes coordinated and independent learning, effectively distinguishing between global and local objectives, leading to faster training convergence than existing methods. Further, non-stationarity arising from independent learning is addressed through a reward backstepping mechanism, while a concatenated action-space reduction strategy reduces the computational complexity of centralized critic training. A position-timing (POST) sequence strategy is proposed to enhance resilience against communication failures by facilitating pseudo co-ordination among agents. Simulation results demonstrate fewer training iterations and greater computational efficiency than benchmark methods. In future, the framework will incorporate more controllable devices and integrate demand response strategies for distribution networks with high DER penetration, unbalanced conditions, and meshed structures.

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